Railtrack Company Code of Practice

METHODOLOGY FOR THE DEMONSTRATION OF COMPATIBILITY WITH TPWS TRACKSIDE EQUIPMENT

ENDORSEMENT & AUTHORISATION

Endorsed by:

....

A Simmons, Professional Head of Signal Engineering

Accepted for issue by:

....

P McAleer, Acting Head of Business Standards

Authorised by:

....

A McNaughton, Chief Engineer

ISSUE RECORD

ISSUE	DATE	COMMENTS
1	February 2003	New Code of Practice

IMPLEMENTATION

The provisions of this code of practice may be used from the date of issue.

DISCLAIMER

SUPPLY

Copies of documents are available electronically, within Railtrack's organisation. Hard copies of this document will be available to Railtrack staff on request to the Document Controller and to other organisations from Technical Indexes Ltd (01344 404409).

Published & Issued by **Railtrack PLC (part of the Network Rail group of companies)**
Railtrack House
Euston Square
LONDON
NW1 2EE

Contents

1. Introduction

1.1 Purpose

The purpose of this document is to provide a methodology to demonstrate compatibility with Train Protection and Warning System (TPWS) trackside equipment on the a.c. and d.c. railways on Railtrack controlled infrastructure.

1.2 Scope

This document describes a method by which a train's compatibility with TPWS used on Railtrack's a.c. and d.c. electrified infrastructure can be assessed. The scope of this document is limited to electromagnetic interference (EMI) emanating from trains and its effect on TPWS equipment installed on the track. It does not consider the effect of train EMI on trainborne TPWS mounted on that train, or on adjacent trains.

2. Source Documents

2.1 Group Standards

1. GE/RT8030: Requirements for a Train Protection and Warning System ` Issue 1, August 2000.

2.2 Previous Safety Case documentation

As TPWS is currently at the installation stage, there are no safety cases, to date, which consider TPWS equipment.

2.3 Other Documentation

2. 'Redifon MEL TPWS Track Sub-System: Technical Characteristics for EMC Assessments', AEAT ref. ESID9004 issue 1.0, dated January 2000.

3. 'TPWS – Rolling Stock Test Strategy', AEAT ref. ESID9010 issue 2.0, dated July 1999.

4. 'TPWS – Results of T&RS Testing', AEAT ref. ESID9066 issue 1.0, dated January 2000.

3. Infrastructure Assumptions

TPWS is required to be installed and maintained to specified standards. Since it is installed as a stand-alone system, the condition of the infrastructure has no bearing upon this analysis.

4. Description of Equipment

4.1 Overview

TPWS has been developed for national implementation in the UK as a means to reduce the number of and minimise the potential consequences of:

- Signals passed at danger (SPADs)
- Excessive train speeds at Permanent Speed Restrictions
- Collisions between trains and buffer stops

To meet this requirement, TPWS provides the functions of a Train Stop System (TSS) and an Overspeed Sensor System (OSS). An emergency brake application will occur on a train fitted with operational TPWS equipment if it passes over an active TSS in the correct direction at any speed, or an active OSS in the correct direction at a speed in excess of a set speed.

If a train is travelling in the 'opposite' direction on a bi-directional line, for example, the trainborne TPWS will ignore signals from the Normal Direction TSS and OSS. An emergency brake application will only occur if the train passes the Opposite Direction TSS and OSS at sufficient speed.

TPWS works alongside the Automatic Warning System (AWS), and does not affect the functionality of AWS. The TPWS vehicle sub-system combines AWS and TPWS system functionality [2].

4.2 Principles of operation

The operation of TPWS is achieved by the transmission of un-modulated tones from transmitter loops located within the four foot. The transmitted tones are received by an antenna located on the underside of a train, in proximity to the operational driving cab. No information is transmitted from the vehicle sub-system to the track sub-system [2].

Both the TSS and OSS functions require two discrete frequency tones to be detected by a train passing over two transmitter loops. For the TSS function, the loops are in close proximity; for the OSS function, the loops are placed a pre-arranged distance apart, where the distance is a measure of the maximum permitted speed. These loops can be energized continuously for protecting a line speed restriction. Alternatively, if protection of a signal is required, the track loops are energised when the corresponding signal has been placed at danger.

In both cases, the first loop 'arms' TPWS in anticipation of a control function and the second loop 'triggers' TPWS to execute the emergency brake function providing both events occur within a short period of time from each other. A pre-set timer within the TPWS trainborne equipment restricts the length of this period [1, 2].

A total of six discrete frequencies are used by TPWS, indicated in Table 1 below, although only one tone frequency is transmitted by each transmitter loop.

Function	Arming Frequency (kHz)	Trigger Frequency (kHz)
Trainstop Sensor (Normal Direction*)	66.25	65.25
Trainstop Sensor (Opposite Direction**)	66.75	65.75
Overspeed Sensor (Normal Direction*)	64.25	65.25
Overspeed Sensor (Opposite Direction**)	64.75	65.75

Table 1: Nominal Tone Frequencies

*Normal Direction
On a line for which signalling is provided for one direction only, 'Normal' applies to the signalled direction. On a bi-directional line, 'Normal' applies to one direction (which may be chosen arbitrarily).

**Opposite Direction
On bi-directional lines Opposite Direction applies to the direction opposite to 'Normal' [1, 2].

4.3 TPWS failure detection

To facilitate remote detection of localised equipment failures a Failure Detection signal is generated automatically and presented on the Signalman's panel. This signal is produced whilst the local signal aspect is at danger and when the track loop appears as an incorrectly terminated transmission line, i.e. when power is reflected back into the transmitter which exceeds 25% of the forward power [3].

5. Susceptibility Analysis

At present there are no defined limits regarding the susceptibility of TPWS to EMC. Therefore testing is required to determine whether interference from rolling stock will affect the correct operation of TPWS trackside equipment. Potential faults which could affect TPWS trackside equipment are described below.

5.1 Susceptibility of trackside equipment

This section considers the possible failure or spurious operation of TPWS trackside equipment due to EMI from trainborne electrical equipment and control systems. The susceptibility of the TPWS power supply is not considered here as it is assumed to have been designed to operate correctly in a railway environment.

Line Filters, Traction Transformers and Radar Units are potential sources of interference, which could cause the monitoring circuit of the TPWS track loop to falsely indicate a fault condition. Since the fault detection circuit on the track loop has a time threshold of 3 seconds, this will only be an issue for trains standing or moving at a very slow speed over the loop. Sources of interference related to traction currents will not be present when the train is stationary.

To ensure compatibility a low speed test should be undertaken as described below. Designers need to be aware of the operating frequency of TPWS and ensure sufficient shielding to minimise emissions at these frequencies [3].

5.1.1 Typical test set-up

A typical test set up is illustrated in figure 1 below. A TPWS power supply/signaling interface module and a train-stop module are mounted in a portable case and used to supply a single track mounted antenna. The second output of the train-stop module is connected to a resistor dummy load. The case includes test connection points for measuring the voltage at the output of the transmitter, a main power supply switch and a switch simulating the input from the signalling system to switch the transmitter output on or off.

The feeder cable, which must be a minimum of 20m in length, is fully laid out alongside the track. The TPWS equipment is also connected to a designated earth point to prevent the development of hazardous touch potentials.

A portable petrol generator can provide the 110V power supply required by the TPWS equipment, if necessary.

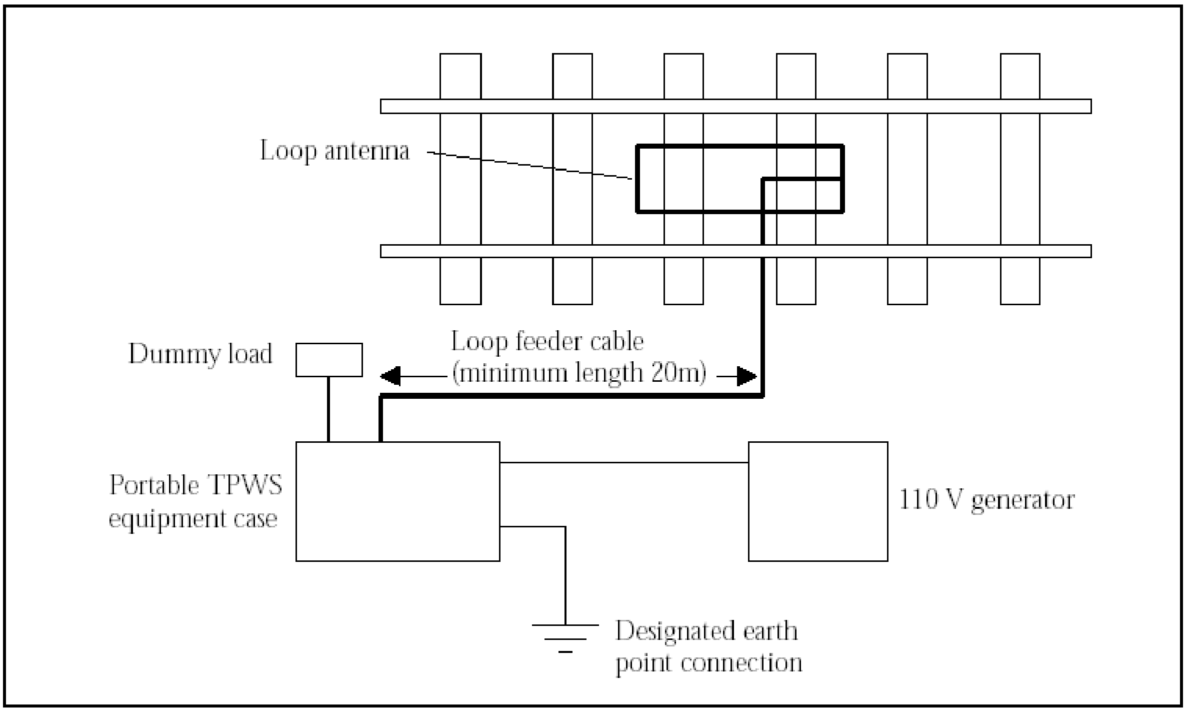

Figure 1: TPWS test equipment arrangement.

The test equipment can be used to perform Pass/Fail and EMI Level tests [4].

5.1.2 Pass/fail test

The aim of the pass/fail test is to determine whether or not interference generated by a train is sufficient to cause a failure of the TPWS equipment. The tests are performed with the track loop transmitting and all instrumentation removed to ensure that interference would not be introduced via the test connections. The results of the tests can be determined from the status of the TPWS power supply module fault indication LED – the test being failed if the LED is illuminated [4].

5.1.3 EMI level test

The aim of the EMI level test is to measure the interference level at the TPWS transmitter output so that the margin can be determined between that and the level required to cause a fault indication. The measurement is made using a storage oscilloscope connected to the test connection points on the portable TPWS equipment case; the transmitter should be switched off during the tests although the TPWS equipment is energised. The voltage waveform at the terminals is to be captured during the interference-generating event and can be saved to disk.

As stated previously, the fault detection circuit of the transmitter will register a fault if greater than 25 % of the forward power is reflected back to the transmitter [4].

6. Methodology for Demonstration of Compatability

As there are no defined limits regarding the susceptibility of TPWS to EMC, testing is required to determine whether interference from rolling stock will affect the correct operation of TPWS. Examples of test methodologies that have been utilised to demonstrate the susceptibility of TPWS trackside equipment are described in section 5. The test arrangement can also be used to determine operating margins for trains running on TPWS fitted routes.

APPENDIX A: PARAMETER VALUES

- TPWS Nominal Tone Frequencies:

Function	Arming Frequency (kHz)	Trigger Frequency (kHz)
Trainstop Sensor (Normal Direction*)	66.25	65.25
Trainstop Sensor (Opposite Direction**)	66.75	65.75
Overspeed Sensor (Normal Direction*)	64.25	65.25
Overspeed Sensor (Opposite Direction**)	64.75	65.75

*Normal Direction
On a line for which signalling is provided for one direction only, 'Normal' applies to the signalled direction. On a bi-directional line, 'Normal' applies to one direction (which may be chosen arbitrarily).

**Opposite Direction
On bi-directional lines Opposite Direction applies to the direction opposite to 'Normal'.

- Power Supply Required by TPWS: 110V

- Time Threshold of TPWS Fault Detection Circuit: 3 seconds

- Minimum Permissible Loop Feeder Cabling Length: 20m

- Maximum Permissible Loop Feeder Cabling Length: 500m

APPENDIX B: ASSUMPTIONS

Configuration

TPWS trackside and trainborne equipment is configured correctly.

Supporting References

1. GE/RT8030: Requirements for a Train Protection and Warning System (TPWS), Issue 1, August 2000.

2. 'Redifon MEL TPWS Track Sub-System: Technical Characteristics for EMC Assessments', AEAT ref. ESID9004 issue 1.0, dated January 2000.

3. 'TPWS – Rolling Stock Test Strategy', AEAT ref. ESID9010 issue 2.0, dated July 1999.

4. 'TPWS – Results of T&RS Testing', AEAT ref. ESID9066 issue 1.0, dated January 2000.

Failure modes

The fault detection circuit of the transmitter will register a fault if greater than 25 % of the forward power is reflected back to the transmitter.

APPENDIX C: HAZARDS

Right Side Failure

A right side failure will result if electromagnetic interference from electrical equipment and control systems cause the monitoring circuit of the TPWS track loop to falsely indicate a fault condition.

Wrong Side Failure

A wrong side failure will result if trains (on adjacent lines to a train fitted with TPWS) produce flux levels at or near TPWS operating frequencies, which are sufficient to prevent the TPWS trainborne aerial from detecting the tone frequency from a trackside loop.

APPENDIX D: MITIGATING FACTORS

No mitigating factors are identified within this document.

APPENDIX E: SOURCES OF TECHNICAL WORK

The technical work contained in this document is taken from testing carried out by AEA Technology Rail.

The traction & rolling stock tests were carried out on various trains.

RAILTRACK

BRIEFING NOTE

RT/E/C/50001 to RT/E/C/50019 **Issue 1**

Methodology for the demonstration of electrical compatibility between rolling stock and infrastructure

Issue Date: February 2003

Background / Synopsis

This suite of Codes of Practice is intended to assist train-builders, TOCs, and ROSCOs in the production of Route Acceptance Safety Cases for electric trains, by describing a methodology for demonstrating compatibility with EE&CS systems on Railtrack's electrified routes.

The electrical susceptibility of the key items of infrastructure is described and characterised, and recommended methods for demonstrating compatibility between rolling stock and infrastructure are given.

Key Changes / Compliance

This is a new suite of documents. As the documents are Codes of Practice, compliance is not mandatory.

Implementation

The information contained in the documents may be used with immediate effect. The various documents in the suite will be published progressively during 2003.

Of Interest To:

Authors of Route Acceptance Safety Cases.

Name and Title of Author

David Wilkinson
Technical & Safety Advisor